

Addendum — The Low-Stakes Outburst: When Magnitude Tracks Disposition, Not Threat

Appends to: "The Neurochemistry of an Acute Social-Threat Stress Response: From Interruption to Sigh of Relief." Field observation, anonymized. Node 0 corpus.

Observed specimen

A subject (unrelated to Node 0; a stranger observed in a public setting) exited a building mid-phone-call, headphones on, handheld game console in hand, and delivered a sustained, high-volume confrontation sequence — *"shut up, listen to me... shut the fuck up, listen to me"* — in the full register of a floor-reclaiming outburst: elevated volume, imperative commands, aggressive prosody. All the surface markers of the acute social-threat response described in the main paper.

On extended observation, the *content* of the dispute was revealed: **headphone brand preference**. The subject had lost one pair, acquired a better pair, and

was defending this position at length and at maximum intensity. That was the entire stakes.

Why this matters: it falsifies a hidden assumption in the main paper

The original paper implicitly modeled a **monotonic coupling between outburst magnitude and threat magnitude** — interruption registers as social threat → amygdala appraisal → SAM discharge → outburst, with the size of the response tracking the size of the provocation. The Node 0 resignation example fit that model cleanly (a genuinely consequential stressor produced a proportionate response).

This specimen **breaks the coupling**. Identical physiological theater — same volume, same imperatives, same dominance signalling — was deployed over a stimulus with essentially zero objective stakes. The response magnitude did **not** track the threat. Therefore, the intensity of an acute social-threat response is **not a reliable readout of the provocation's severity**.

The corrected model: magnitude is a joint function, and disposition can dominate

The response magnitude is a product of at least two multiplicative terms:

1. **Appraised threat** — how threatening the amygdala/prefrontal appraisal judges the situation (main paper, Step 0).
2. **Dispositional gain** — the individual's trait-level reactivity: trait anger/hostility, baseline vagal tone (low resting HRV = weaker brake), stress sensitization, testosterone/dominance orientation, COMT/MAOA genotype, and — critically — **learned reinforcement history** (main paper, Steps 6–7).

When dispositional gain is high, a **trivial trigger can drive a maximal response**, because the appraisal step is being amplified by a large multiplier. What looks like a large threat is often a large *gain* applied to a small one. The headphone subject is the limiting case: near-zero threat, very high gain, maximal output.

This connects directly to the main paper's **reinforcement loop (Step 6)**. A person for whom the confrontation-outburst has been *repeatedly negatively reinforced* — it reliably ends the aversive state and produces the clean parasympathetic/dopaminergic relief — will have a **lowered threshold and raised gain** for deploying it. Over enough cycles, the outburst stops being a proportionate response to threat and becomes a **default affect-regulation strategy**, fired at

low-stakes triggers because it *works* (it reliably produces relief), not because the trigger warrants it. The trivial-content outburst is what an entrenched reinforcement loop looks like from the outside.

Diagnostic implication

This gives an observable heuristic:

- **Outburst magnitude proportionate to stakes** → likely a situational SAM response; the trigger is doing the work.
- **Outburst magnitude wildly disproportionate to stakes** (maximal response, trivial content) → the *disposition* is doing the work, not the situation. This is the signature of high dispositional gain and/or an entrenched reinforcement loop. The content is close to irrelevant; almost anything would have served as the trigger.

The corollary for self-observation: if you notice your *own* response magnitude decoupling from stakes — going to maximum over something that objectively doesn't warrant it — that is the diagnostic sign that the reinforcement loop (not the provocation) is in the driver's seat, and the loop-breaking interventions from the main paper (Recommendations) are indicated regardless of how justified the specific trigger feels in the moment.

Caveats

- **Single observation, external, no context beyond overheard content.** The behavior is logged as observed; the interpretation (high dispositional gain vs. e.g. performative/gaming trash-talk, a bit, or a genuinely consequential subtext not audible) is a hypothesis, not a confirmed reading. The specimen is real as *behavior observed*; the mechanistic attribution is provisional.
- **Provenance gate.** This is an anonymized behavioral specimen only. No identifying detail, no image of the subject, who did not consent to observation. The behavior belongs in the corpus; the person does not.
- The multiplicative appraisal $\times$ gain framing is a simplification of a more distributed process, offered as a working model, not a measured parameterization.